Revisiting Isolated DC-DC Converters-III

Introduction

In the realm of modern power electronics, particularly for critical applications like electric vehicle charging infrastructure, the efficient and reliable conversion of DC power is paramount. Isolated DC-DC converters play a crucial role by providing galvanic isolation between the input and output, enhancing safety and allowing for flexible grounding schemes. This lecture delves into advanced isolated DC-DC converter topologies, specifically the full-bridge and half-bridge converters, building upon the foundational understanding of forward and push-pull converters. These topologies address some of the limitations of their predecessors, offering improved performance characteristics such as better magnetic core utilization and reduced voltage stress on switching devices, which are vital for high-power and high-frequency applications.

Understanding these converters is essential for designing robust and compact power conversion systems. By exploring their operational principles, advantages, and component sizing methodologies, we gain insights into how these circuits manage power flow, minimize losses, and ensure stable output under varying load conditions. The full-bridge and half-bridge converters represent significant advancements in power converter design, enabling higher power densities and greater efficiency, which are critical for the evolving demands of energy systems.

Core Concepts

Term	Definition	Significance
`Full-Bridge Converter`	An isolated DC-DC converter topology utilizing four switching devices arranged in a full-bridge configuration on the primary side of a transformer.	Offers full magnetic core utilization and reduced voltage stress on switches, suitable for higher power applications.
`Half-Bridge Converter`	An isolated DC-DC converter topology using two switching devices and two series-connected capacitors on the primary side of a transformer.	Requires fewer switches than a full-bridge but utilizes only half of the input DC bus voltage across the transformer primary.

Term	Definition	Significance
`Flux Utilization`	The extent to which the magnetic core of a transformer is utilized, typically measured by the range of flux density (e.g., from -Bmax to +Bmax).	Maximizing flux utilization allows for a smaller, more efficient transformer core for a given power level.
`Voltage Kickback`	High voltage spikes generated across switching devices due to the leakage inductance of transformer windings during turn-off.	A significant concern in some topologies (e.g., push-pull) that necessitates snubber circuits or careful design to protect switches.
`Ripple Frequency (2fsw)`	The frequency of the ripple component in the output inductor current and capacitor voltage, which is twice the switching frequency.	A higher ripple frequency allows for smaller output inductors and capacitors, leading to more compact and cost-effective designs.
`Volt-Second Balance`	The principle stating that the average voltage across an inductor over one switching period must be zero in steady-state operation.	Fundamental for deriving the input-output voltage relationship (gain) of DC-DC converters.
`MOSFET`	Metal-Oxide-Semiconductor Field-Effect Transistor, a type of semiconductor device used as a switch in power converters.	Preferred for high-frequency switching applications due to faster switching speeds and lower conduction losses compared to IGBTs.
`Isolated DC-DC Converter`	A power converter that provides galvanic isolation between its input and output using a transformer.	Essential for safety, noise reduction, and flexible grounding in many power supply applications, including charging infrastructure.

Detailed Analysis

1. Evolution of Isolated DC-DC Converters

The journey through isolated DC-DC converters often begins with simpler topologies like the forward converter. While effective, the forward converter typically utilizes the magnetic core only in one direction (e.g., 0 to Bmax), leading to underutilization of the core material. Variations like the forward converter with a unit demagnetizing winding and the dual-switch forward converter aimed to improve performance but still had limitations.

The push-pull converter marked an advancement by achieving full core utilization, allowing the flux to swing from -Bmax to +Bmax. This bidirectional flux swing means that for the same voltage, number of turns, and frequency, the core area can be reduced by half, leading to a more compact transformer. However, the push-pull converter introduced its own set of challenges, notably the issue of voltage kickback due to leakage inductance and the requirement for switching devices to be rated for twice the input voltage (2Vin), which increases component cost and complexity. These limitations paved the way for more sophisticated topologies like the full-bridge converter.

2. Full-Bridge DC-DC Converter Operation

The full-bridge converter is a robust isolated DC-DC topology that addresses many of the shortcomings of its predecessors. It consists of two identical half-bridges on the primary side, totaling four switching devices (typically MOSFETs for high-frequency operation). The primary winding of the transformer is connected between the poles of these two half-bridges. The secondary side typically employs a full-bridge rectifier followed by an LC filter to smooth the output.

The operation can be understood through distinct switching intervals:

- **Interval 1: S1 and S4 ON (DTs/2 period)**
- Switches S1 and S4 are turned on simultaneously.
- This applies the full input voltage (Vin) across the primary winding of the transformer.
- The primary voltage (Vp) is equal to Vin.
- Due to the positive voltage across the primary, the magnetic flux in the core starts to rise linearly ($V = N * d\Phi/dt$).
- On the secondary side, the voltage becomes n * Vin (where n = Ns/Np).
- Diode DR1 becomes forward biased, and DR2 becomes reverse biased.

- The voltage across the output inductor (VL) is n * Vin - Vo, causing the inductor current (IL) to rise with a positive slope.
- **Interval 2: All Switches OFF (1-D)Ts/2 period - Dead Time**
- After S1 and S4 turn off, all four primary switches (S1, S2, S3, S4) are off.
- During this period, the output capacitances of the switches (C1, C2, C3, C4) come into play. Assuming identical switches, these capacitances equally divide the input voltage, resulting in Vin/2 across each half-bridge.
- Crucially, the voltage across the primary winding (Vp) becomes zero.
- Since Vp is zero, the rate of change of flux (dΦ/dt) is zero, meaning the magnetic flux in the core remains constant at its peak value (Φm).
- On the secondary side, the inductor current (IL), which cannot change instantaneously, finds a path through both DR1 and DR2, causing both diodes to become forward biased. This effectively short-circuits the secondary winding.
- The voltage across the output inductor (VL) becomes -Vo, causing the inductor current to fall with a negative slope.
- **Interval 3: S2 and S3 ON (DTs/2 period)**
- Switches S2 and S3 are turned on simultaneously.
- This applies a negative input voltage (-Vin) across the primary winding.
- The primary voltage (Vp) is equal to -Vin.
- The magnetic flux in the core starts to fall linearly, moving from +Φm towards -Φm.
- On the secondary side, the voltage becomes -n * Vin.
- Diode DR1 becomes reverse biased, and DR2 becomes forward biased.
- The voltage across the output inductor (VL) is n * Vin - Vo (in magnitude), causing the inductor current to rise again.
- **Interval 4: All Switches OFF (1-D)Ts/2 period - Dead Time**
- Similar to Interval 2, all switches are off, Vp is zero, flux is constant, and VL is -Vo.

Advantages of the Full-Bridge Converter:

* **Full Core Utilization:** The flux in the transformer core swings from -Φm to +Φm, ensuring optimal utilization of the magnetic material.
* **Reduced Switch Voltage Stress:** Each primary switch only needs to block the input voltage (Vin), unlike the push-pull converter where switches block 2Vin. This allows for the use of lower voltage-rated, and thus often cheaper and more efficient, switches.

* **High Output Ripple Frequency:** The output inductor current and capacitor voltage ripple components occur at twice the switching frequency (2fsw). This allows for smaller inductance and capacitance values for the output filter, leading to a more compact and responsive design.

3. Sizing Components: Inductor and Capacitor

Accurate sizing of the output inductor (L) and capacitor (C) is crucial for meeting output ripple specifications and ensuring stable operation.

- **Output Voltage Derivation (Volt-Second Balance):**
  Applying volt-second balance across the output inductor over a Ts/2 period (since the inductor current repeats every Ts/2):
  (n * Vin - Vo) * (DTs/2) + (-Vo) * (1-D)Ts/2 = 0
  Simplifying this equation yields the output voltage:
  Vo = n * Vin * D
  This relationship is similar to that of the forward and push-pull converters, highlighting the fundamental DC conversion ratio.
- **Inductor Sizing (L):**
  During the (1-D)Ts/2 period, the voltage across the inductor is -Vo. The change in inductor current (ΔIL) during this period can be related to L:
  VL = L * (dIL/dt)
  -Vo = L * (ΔIL / ((1-D)Ts/2))
  Rearranging for L:
  L = (Vo * (1-D) * Ts) / (2 * ΔIL)
  Since Ts = 1/fsw, and the ripple frequency is 2fsw, we can write:
  L = (Vo * (1-D)) / (2 * fsw * ΔIL)
  Here, ΔIL is the peak-to-peak inductor current ripple, typically specified as a percentage (e.g., 5-10%) of the average output current (Io).
- **Capacitor Sizing (C):**
  The output capacitor filters the ripple current from the inductor. The ripple component of the inductor current (ΔIL) flows through the capacitor. The voltage ripple across the capacitor (ΔVo) is related to the charge stored/released.
  The charge (ΔQ) flowing into/out of the capacitor can be approximated by the area of the triangular ripple current waveform. For a ripple frequency of 2fsw, the period for one half-cycle of ripple is Ts/4. The peak ripple current is ΔIL/2.
  ΔQ = (1/2) * (ΔIL/2) * (Ts/4) = (ΔIL * Ts) / 16
  Since ΔQ = C * ΔVo:
  C = ΔQ / ΔVo = (ΔIL * Ts) / (16 * ΔVo)

Substituting Ts = 1/fsw:
C = ΔIL / (16 * fsw * ΔVo)
However, the lecture provides a slightly different formula, likely based on a different interpretation of the ripple waveform or a simplified approximation:
C = ΔIL / (8 * fsw * ΔVo)
This formula implies that the capacitor needs to be sized for the ripple current at twice the switching frequency (2fsw), where ΔVo is the specified output voltage ripple.

4. Half-Bridge DC-DC Converter Operation

The half-bridge converter is a variation of the isolated DC-DC topology that offers a trade-off between component count and voltage utilization. Instead of two half-bridges with switches, it uses one half-bridge with two switches (S1, S2) and another half-bridge composed of two series-connected capacitors (C3, C4) across the input voltage (Vin). The midpoint of these capacitors provides a stable Vin/2 potential. The primary winding of the transformer is connected between the midpoint of the switches and the midpoint of the capacitors.

- **Operation:**
- When S1 is ON, the voltage across the primary winding is Vin/2.
- When S2 is ON, the voltage across the primary winding is -Vin/2.
- Similar to the full-bridge, there are dead times where both switches are off, and the primary voltage is zero.
- The secondary side rectification and filtering operate identically to the full-bridge, resulting in a 2fsw ripple frequency for L and C.
- **Output Voltage Derivation:**
  Applying volt-second balance, considering the primary voltage is now Vin/2:
  Vo = n * (Vin/2) * D
- **Comparison with Full-Bridge:**
- **Advantages:** Requires only two primary switches, reducing complexity and cost compared to the full-bridge's four switches.
- **Disadvantages:** The primary winding only sees half of the input voltage (Vin/2). This means that for the same output power, the primary current will be higher, or the transformer turns ratio (n) needs to be adjusted, potentially leading to a larger transformer or higher conduction losses. It effectively reduces the DC bus utilization by half.
- **Similarities:** Both topologies achieve full core utilization and benefit from a 2fsw ripple frequency for the output filter components.

Key Takeaways

- [ ] Full-bridge converters utilize four switches (typically MOSFETs) arranged in two half-bridges on the primary side.
- [ ] Full-bridge converters achieve full magnetic core utilization by driving flux from -Bmax to +Bmax.
- [ ] The voltage stress on primary switches in a full-bridge converter is limited to Vin, an advantage over push-pull converters.
- [ ] The output inductor and capacitor in both full-bridge and half-bridge converters are sized for twice the switching frequency (2fsw), allowing for smaller filter components.
- [ ] The output voltage of a full-bridge converter is given by Vo = n * Vin * D, derived using volt-second balance.
- [ ] Half-bridge converters use two switches and two capacitors on the primary side, applying Vin/2 across the transformer primary.
- [ ] The output voltage of a half-bridge converter is Vo = n * (Vin/2) * D, indicating reduced DC bus utilization compared to the full-bridge.
- [ ] Component sizing for L and C depends on the desired ripple current (ΔIL) and ripple voltage (ΔVo), and the 2fsw ripple frequency.

Conclusion

The full-bridge and half-bridge isolated DC-DC converters represent sophisticated solutions for power conversion, each offering distinct advantages and trade-offs. The full-bridge converter stands out for its excellent magnetic core utilization and reduced voltage stress on switching devices, making it a preferred choice for high-power applications where efficiency and component reliability are paramount. Its ability to fully utilize the transformer core and limit switch voltage to Vin contributes to more compact and robust designs.

The half-bridge converter, while requiring fewer switching devices, sacrifices some DC bus utilization, resulting in a lower output voltage for a given turns ratio or requiring higher primary currents. However, its reduced component count can be advantageous in applications where cost and simplicity are critical, and the power levels are moderate. Both topologies share the benefit of a 2fsw ripple frequency, which allows for smaller and more efficient output filter components. The choice between these converters ultimately depends on the specific application requirements, including power level, voltage stress tolerance, cost constraints, and desired power density, underscoring the importance of a thorough understanding of their operational characteristics for effective power system design.